

# Mario Casas Donjuan

Monterrey, N.L., Mexico

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## PROFESSIONAL SUMMARY

B.S. in Robotics and Digital Systems Engineering undergraduate working across embedded electronics, control systems, and software automation. Develops avionics hardware and control logic for aerospace applications, Java control systems for competitive robotics (three World Championship qualifications), and Python services using FastAPI, computer vision, and machine learning. Delivered a workflow automation system for an internal SAP team under a six-week constraint. Skilled in PCB design, Linux administration, and virtualization (Proxmox VE, Docker) from running a self-hosted home lab. Passionate about software systems and their optimization, aiming to improve their efficiency and connection with the real world.

## EDUCATION & CERTIFICATIONS

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| <b>Tecnológico de Monterrey</b><br><i>B.S. in Robotics and Digital Systems Engineering (IRS)</i><br>Grade Average: 88/100 (Preliminary)                                                                                                                                                      | Monterrey, NL<br><i>Aug. 2025 – Jun. 2029 (Exp.)</i> |
| <b>Technical Certifications</b><br><i>CS50 (Harvard University)   Microsoft Azure Fundamentals AZ-900 (In Progress)</i>                                                                                                                                                                      | Online<br><i>2025 – Present</i>                      |
| <b>Prepa Tec Eugenio Garza Lagüera</b><br><i>Multicultural Baccalaureate (IB Program)</i><br>Grade Average: 88/100   Academic Distinctions: Honors Program, Global Citizenship Stole, LiFE STEAM Stole.<br>Focus on Object-Oriented Programming (Java, Python, C++) and Global Competencies. | Monterrey, NL<br><i>Aug. 2022 – Jun. 2025</i>        |
| <b>UANL School of Music (FAMUS)</b><br><i>Music Theory and Instrumental Execution (Cello)</i><br>5-year rigorous program focused on Solfège, Harmony, and Technical Execution.                                                                                                               | Monterrey, NL<br><i>2017 – 2022</i>                  |

## TECHNICAL EXPERIENCE

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| <b>SAP – Talent Lab Program</b><br><i>Software Intern, Workflow Automation</i><br>Automated a manual international call scheduling workflow for an internal support team using Power Automate, SharePoint, and Microsoft Teams.<br>Built time zone matching logic computing UTC overlap across regions, including scheduling windows that cross midnight.<br>Delivered Adaptive Card notifications routed to the appropriate Teams channel with dynamic @mentions.<br>Authored complete technical documentation enabling the team to replicate, maintain, and extend the system after handoff.<br>Gathered and iterated on requirements directly with stakeholders under a 6-week delivery constraint. | Monterrey, NL<br><i>Jun. 2026 – Jul. 2026</i> |
| <b>Monterrey Rocket Lab (MRL)</b><br><i>Avionics Engineer</i><br>Develop and integrate electronic systems and control logic for aerospace applications.<br>Build telemetry systems for real-time data acquisition and transmission during flight.                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Monterrey, NL<br><i>Sep. 2025 – Present</i>   |
| <b>FIRST Robotics Competition – Team Voltec 6647</b><br><i>Programming Mentor</i><br>Supervise software architecture and mentor junior developers on OOP principles and best practices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Monterrey, NL<br><i>Jun. 2025 – Present</i>   |
| <b>FIRST Robotics Competition – Team Voltec 6647</b><br><i>Team Leader &amp; Main Driver</i><br>Developed Object-Oriented Java code for autonomous and teleoperated control of 60 kg industrial robots.<br>Implemented PID control algorithms and computer vision for automatic positioning.<br>Proposed and implemented cross-disciplinary solutions integrating software, electronics, and mechanical subsystems.<br>Achievements: Regional Champion, multiple technical awards, and World Championship qualification for 3 years.                                                                                                                                                                   | Monterrey, NL<br><i>Aug. 2022 – Jun. 2025</i> |
| <b>Escuela Voltec</b><br><i>Java Programming Mentor</i><br>Designed course material and delivered a structured OOP program for middle school students.<br>Guided students in understanding programming logic and connecting code with physical robotic systems.                                                                                                                                                                                                                                                                                                                                                                                                                                        | Monterrey, NL<br><i>2024</i>                  |

## PROJECTS

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- Home Lab Server & Portfolio** | *Proxmox VE, Linux, Docker, Networking, React* Jan. 2025 – Present
- Deployed a bare-metal server with Proxmox VE to manage Linux virtual machines, centralizing 24/7 service hosting.
  - Configured Docker containers and secure network tunnels (VLANs, Tailscale) for public access without exposing local ports.
  - Developed a personal portfolio website using React + Vite with automated Git deployment.
- Industrial IoT Startup Development** | *Systems Architecture, B2B* Apr. 2026 – Present
- Designed software infrastructure and data transmission pipeline for a B2B model, enabling precise remote cold-chain monitoring.
- RepLab – Exercise Form Analysis** | *Python, FastAPI, OpenCV, MediaPipe, SQLite* May 2026
- Built a computer-vision pipeline segmenting exercise repetitions by tracking direction changes in a 5-frame rolling window, extracting three annotated keyframes per rep and computing a 0–100 form score.
  - Served the pipeline through a FastAPI backend offloading CPU-bound inference to a thread executor, integrating the Google Gemini API for structured feedback and persisting sessions to SQLite.
  - Built in a one-day hackathon (HackDays 2026).
- Magnetic Braking Computational Simulation** | *MATLAB, C++* May 2026
- Engineered scripts to compute deceleration using the Biot-Savart law, producing accurate physical prediction models.
- MichiBeam Automated System** | *ESP32, C++, 3D Printing, CAD* Feb. 2026 – Mar. 2026
- Engineered an autonomous hardware-software integration utilizing microcontrollers and servomotors.
  - Designed physical components using CAD and applied mathematical modeling for precise control logic.
- Airline Inventory Prediction (HackMTY)** | *Python, scikit-learn, FastAPI, SQLite, React* Oct. 2025
- Trained a RandomForestRegressor inside a scikit-learn pipeline (ColumnTransformer, OneHotEncoder with unseen-category handling), serialized with joblib and served through a live inference endpoint.
  - Designed a two-table SQLite schema and the raw-SQL data access layer powering an 11-endpoint REST API.
- Flipply (HackHarvard)** | *Python, FastAPI, Vertex AI, eBay API, React Native* Oct. 2025
- Contributed to a 5-person team building a mobile app that identifies an item from a photo and prices it against live eBay comparables using a two-stage LLM pipeline.
- Pharmaceutical Management App (IB Project)** | *Java, MySQL, Swing* Aug. 2024 – May 2025
- Engineered a desktop application for pharmaceutical inventory management with custom authentication.
  - Designed and deployed a normalized relational database in MySQL.

## LEADERSHIP & MODEL UNITED NATIONS (MUN)

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- Model United Nations (MUNMX)** Monterrey, NL
- UNODC Committee Chair & Delegate* Nov. 2023 – Nov. 2024
- Moderated formal debate sessions, enforced parliamentary procedure, and evaluated delegates' performance.
- Developed advanced argumentation, negotiation, and public speaking skills while managing crisis scenarios.

## MUSICAL & ARTISTIC TRAJECTORY

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- FAMUS Symphony Orchestra & Guitar Ensemble** UANL
- Principal Cello & Soloist Guitarist* 2017 – 2022
- Led the cello section in orchestral performances and won the Faculty Guitar Contest in 2021 and 2022.
- Cultivated discipline, synchronous teamwork, and artistic sensitivity through classical interpretation.

## SKILLS & LANGUAGES

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- Languages:** Spanish (Native), English (C1 – Advanced), German (A1), French (A2).
- Programming:** Python (Advanced), Java (Advanced), C++ (Intermediate), TypeScript, SQL (MySQL, SQLite), MATLAB, C (Beginner).
- Technologies:** FastAPI, scikit-learn, OpenCV, MediaPipe, Pandas, NumPy, React, Docker, Proxmox VE, Git/GitHub, Linux (Arch/Debian), Tailscale, Power Automate, SharePoint.
- Software:** VS Code, Neovim, Altium Designer, Onshape (CAD), LaTeX, Microsoft Office.
- Soft Skills:** Leadership, Problem-solving, Fast learning, Work under pressure, Analytical thinking.
- Technical Skills:** PCB design and schematic capture, microcontroller and system programming, automation and robotics, algorithms and data structures.

## AWARDS & COMPETITIONS

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- FIRST Robotics Competitions (2022–Present):** Participated in 16 regional/international events; Regional Winner (1st Place) Hermosillo 2024 and 3x World Championship Qualifier.
- RoboChamps 2024:** Semifinalist (Advanced Category).
- Sports:** Table Tennis Representative Team (Prepa Tec, 2022–2025).